

Main Activity

```
package com.zenhealth.janaushadhi
```

```
import android.os.Bundle
```

```
import android.widget.SearchView
```

```
import androidx.appcompat.app.AppCompatActivity
```

```
import androidx.recyclerview.widget.LinearLayoutManager
```

```
import androidx.recyclerview.widget.RecyclerView
```

```
import com.zenhealth.janaushadhi.adapters.MedicineAdapter
```

```
import com.zenhealth.janaushadhi.models.Medicine
```

```
class MainActivity : AppCompatActivity() {
```

```
    private lateinit var recyclerView: RecyclerView
```

```
    private lateinit var adapter: MedicineAdapter
```

```
    private val medicineList = mutableListOf<Medicine>()
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
```

```
        super.onCreate(savedInstanceState)
```

```
        setContentView(R.layout.activity_main)
```

```
        // Mock Data (Matches your Jan Aushadhi app)
```

```
        loadMedicines()
```

```
        recyclerView = findViewById(R.id.medicineRecyclerView)
```

```
        adapter = MedicineAdapter(medicineList)
```

```
        recyclerView.layoutManager = LinearLayoutManager(this)
```

```
        recyclerView.adapter = adapter
```

```
        val searchView = findViewById<SearchView>(R.id.medicineSearch)
```

```

searchView.setOnQueryTextListener(object : SearchView.OnQueryTextListener {
    override fun onQueryTextSubmit(query: String?): Boolean = false
    override fun onQueryTextChange(newText: String?): Boolean {
        adapter.filter(newText)
        return true
    }
})
}

private fun loadMedicines() {
    medicineList.add(Medicine("1", "Paracetamol", "Fever", 1.20, 12.50, "500mg", "Pain
and fever relief"))
    medicineList.add(Medicine("2", "Amoxicillin", "Antibiotic", 2.50, 18.00, "250mg",
"Bacterial infections"))
    medicineList.add(Medicine("3", "Metformin", "Diabetes", 1.50, 15.00, "500mg", "Blood
sugar control"))
    // Add more as needed...
}
}

```

Dependencies

```

dependencies {
    implementation("androidx.recyclerview:recyclerview:1.3.2")
    implementation("com.google.android.material:material:1.11.0")
    implementation("androidx.cardview:cardview:1.0.0")
    implementation("com.github.bumptech.glide:glide:4.16.0") // For images if needed
}

```

Data Model

```
data class Medicine(  
    val id: String,  
    val name: String,  
    val category: String,  
    val genericPrice: Double,  
    val brandedPrice: Double,  
    val composition: String,  
    val description: String  
)
```

Layout (activity_main.xml)

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:orientation="vertical"  
    android:background="#FAFAF9"  
    android:padding="16dp">  
  
    <TextView  
        android:layout_width="wrap_content"  
        android:layout_height="wrap_content"  
        android:text="ZenHealth"  
        android:textColor="#065F46"  
        android:textSize="24sp"  
        android:textStyle="bold" />  
  
    <SearchView  
        android:id="@+id/medicineSearch"
```

```

        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:background="@drawable/search_bg"
        android:queryHint="Search generic medicines..." />

<androidx.recyclerview.widget.RecyclerView
    android:id="@+id/medicineRecyclerView"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_marginTop="16dp" />
</LinearLayout>

```

Summery

1. **Medicine Comparison:** It shows the generic price vs. the branded price side-by-side with a calculated "Savings %" badge.
2. **Real-time Search:** The SearchView filters the list as you type.
3. **Modern UI:** Uses Material Design 3 cards and consistent spacing, matching the "Zen" aesthetic for app.
4. **Database Integration:** You can easily swap the loadMedicines() mock data with a **Firestore** or **Room** database call.